



WIT

**BACHELOR OF SCIENCE (HONS) IN
- APPLIED COMPUTING
- COMPUTER FORENSICS & SECURITY
- ENTERTAINMENT SYSTEMS
- THE INTERNET OF THINGS**

EXAMINATION:

**DISCRETE MATHEMATICS
(COMMON MODULE)
SEMESTER 1 - YEAR 1**

DECEMBER 2021

DURATION: 2 HOURS

**INTERNAL EXAMINERS: DR DENIS FLYNN
DR KIERAN MURPHY**

**DATE: 20 DEC 2021.
TIME: 14.15 PM
VENUE: MAIN HALL**

EXTERNAL EXAMINER: MS MARGARET FINNEGAN

INSTRUCTIONS TO CANDIDATES

- 1. ANSWER ALL QUESTIONS.**
- 2. TOTAL MARKS = 100.**
- 3. EXAM PAPER (5 PAGES) AND FORMULA SHEET (1 PAGE)**

MATERIALS REQUIRED

- 1. NEW MATHEMATICS TABLES.**
- 2. GRAPH PAPER**

WATERFORD INSTITUTE OF TECHNOLOGY

Question 1

- (a) Alice, Bob and Carol are three students that took last year's *Discrete Mathematics* exam. Given propositions

- $A = \text{"Alice passed the exam."}$
- $B = \text{"Bob passed the exam."}$
- $C = \text{"Carol passed the exam."}$

Formalise (i.e., express using logical operators) the following statements

- (i) $\text{"Alice is the only one passing the exam."}$
- (ii) $\text{"Bob passed the exam."}$
- (iii) $\text{"All three students passed the exam."}$
- (iv) $\text{"At least one student, among Alice, Bob and Carol, passed the exam."}$
- (v) $\text{"Exactly one student passed the exam."}$

(10 marks)

- (b) Which of the following are well formed propositional formulas? Justify your answers.

- (i) $\forall pq$
- (ii) $p \neg \neg r$
- (iii) $p \neg \rightarrow (q \wedge q)$
- (iv) $(p \wedge \neg q) \vee (q \neg \rightarrow q)$

(4 marks)

- (c) Consider the functions defined by the following Python code:
(recall that $//$ is integer division, and $**$ is exponentiation (powers).)

```
1 def f(x):  
2     return x ** 2  
3  
4 def g(x):  
5     return x + 1  
6  
7 def h(x):  
8     return x // 2
```

Evaluate the following:

- (i) $f(5)$
- (iii) $h(3)$
- (v) $f(5) + g(7)$
- (ii) $f(g(2))$
- (iv) $f(g(h(1)))$
- (vi) $f(f(f(2)))$

(6 marks)

(Total 20 marks)

Question 2

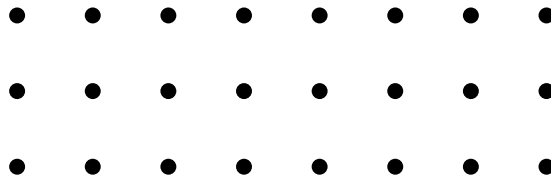
- (a) Use truth tables to determine whether the proposition

$$(\neg p \vee q) \wedge (q \rightarrow (\neg r \wedge \neg p)) \wedge (p \vee r)$$

is satisfiable.

(5 marks)

- (b) Consider the follow diagram, consisting of three rows of eight dots.



How many

- (i) Squares (*Hint: Count 1×1 and 2×2 sized squares separately.*)
- (ii) Rectangles (*Hint: Count cases based on top row of rectangle.*)
- (iii) Right angled triangles

can be drawn using the dots as vertices (corners).

(9 marks)

- (c) Draw the following graphs. Classify each as Eulerian, semi-Eulerian, or neither.

- (i) The wheel graph, W_4 .
- (ii) The complete bipartite graph, $K_{4,2}$.
- (iii) The Peterson graph.

(6 marks)

(Total 20 marks)

Question 3

- (a) Construct sets A and B satisfying the following three properties:

$$A \setminus B = \{4, 9\}, \quad B \setminus A = \{8\}, \quad A \cap B = \{1\}.$$

(2 marks)

- (b) Consider the *Collatz function* with definition and Python implementation

$$f(n) = \begin{cases} 3n + 1 & n \text{ odd} \\ n/2 & n \text{ even} \end{cases}$$

```
1 def f(n):  
2     if n%2==1:  
3         return 3 * n + 1  
4     else:  
5         return n // 2
```

Also implemented in python are the following functions

```
1 def g(n):  
2     return f(f(f(n)))  
3  
4 def h(n):  
5     result = [n]  
6     while n!=1:  
7         n = f(n)  
8         result.append(n)  
9     return result
```

- (i) What is the output of $f(7)$? and $f(10)$?
- (ii) What is the output of $h(12)$?
- (iii) What is the output of $g(10)$?
- (iv) For which values of n does the expression $g(n)==n$ compute to `True`?

(8 marks)

- (c) Let $S = \{1, 2, 3, 4, 5, 6\}$

- (i) How many subsets are there of cardinality 4?
- (ii) How many subsets of S are there? That is, find $|\mathcal{P}(S)|$.
- (iii) How many subsets of cardinality 4 have $\{2, 4, 6\}$ as a subset?
- (iv) How many subsets of cardinality 4 contain at least one prime (2, 3, or 5)?
- (v) How many subsets of cardinality 4 contain exactly one prime?

(10 marks)

Question 4

- (a) Let $A = \{0, 2, 3\}$, $B = \{2, 3\}$, and $C = \{1, 5, 9\}$. Determine which of the following statements are true. Give reasons for your answers.

- | | |
|---------------------------|------------------------------|
| (i) $3 \in A$ | (iv) $\emptyset \subseteq C$ |
| (ii) $\{3\} \in A$ | (v) $\emptyset \in A$ |
| (iii) $\{3\} \subseteq A$ | (vi) $A \subseteq A$ |

(6 marks)

- (b) Let R be the relation on the set $A = \{0, 1, 2, 3\}$ given by

$$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$$

- (i) Represent R using a digraph.
- (ii) Is R reflexive? symmetric? transitive? Justify your answers.
- (iii) Is R irreflexive? Justify your answer.

(2 + 6 + 1 = 9 marks)

- (c) How many shortest lattice paths start at (2,4) and

- (i) end at (20,15)?
- (ii) end at (20,15) and pass through (10,6)?
- (iii) end at (20,15) and avoid (10,6)?

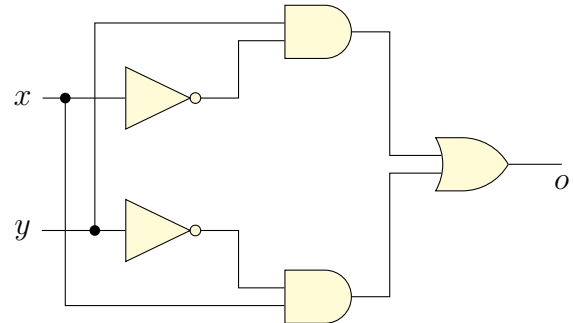
(1 + 2 + 2 = 5 marks)

(Total 20 marks)

Question 5

(a) Consider the following logical circuit with two inputs, and single output.

- (i) Construct a logical expression to represent this circuit.
- (ii) Is there an input case for which the output is on?
- (iii) Hence, construct a logical circuit or equivalent expression, containing three inputs, x , y , and z , for which the output is on when exactly one input is on.



(2 + 2 + 2 = 6 marks)

(b) Consider the function $f : \{1, 2, 3, 4, 5\} \rightarrow \{1, 2, 3, 4\}$ given by the table below:

x	1	2	3	4	5
$f(x)$	3	2	4	1	2

- (i) Is f injective? Explain.
- (ii) Is f surjective? Explain.

(4 marks)

(c) How many 9-bit strings (that is, bit strings of length 9) are there which satisfy each of the following criteria? Explain your answers.

- (i) Start with the sub-string 101.
- (ii) Have weight 5 (i.e., contain exactly five 1's) and start with the sub-string 101.
- (iii) Either start with 101 or end with 11 (or both).
- (iv) Have weight 5, and starts with 101 and ends with 11.

(2 + 2 + 3 + 3 = 10 marks)

(Total 20 marks)

Laws of Logic

Logical Connective	Symbol	Python Operator	Precedence	Logic Gate
Negation (NOT)	$\neg$	<code>not</code>	Highest	
Conjunctive (AND)	$\wedge$	<code>and</code>	Medium	
Disjunctive (OR)	$\vee$	<code>or</code>	Lowest	

Basic Rules of Logic

Commutative Laws

$$p \vee q \Leftrightarrow q \vee p \quad p \wedge q \Leftrightarrow q \wedge p$$

Associative Laws

$$(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r) \quad (p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$$

Distributive Laws

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r) \quad p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

Identity Laws

$$p \vee \mathbf{F} \Leftrightarrow p \quad p \wedge \mathbf{T} \Leftrightarrow p$$

Negation Laws

$$p \wedge (\neg p) \Leftrightarrow \mathbf{F} \quad p \vee (\neg p) \Leftrightarrow \mathbf{T}$$

Idempotent Laws

$$p \vee p \Leftrightarrow p \quad p \wedge p \Leftrightarrow p$$

Null Laws

$$p \wedge \mathbf{F} \Leftrightarrow \mathbf{F} \quad p \vee \mathbf{T} \Leftrightarrow \mathbf{T}$$

Absorption Laws

$$p \wedge (p \vee q) \Leftrightarrow p \quad p \vee (p \wedge q) \Leftrightarrow p$$

DeMorgan's Laws

$$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q \quad \neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$$

Involution Law

$$\neg(\neg p) \Leftrightarrow p$$

Implications and Equivalences

Detachment (Modus Ponens)

$$(p \rightarrow q) \wedge p \Rightarrow q$$

Indirect Reasoning (Modus Tollens)

$$(p \rightarrow q) \wedge \neg q \Rightarrow \neg p$$

Disjunctive Addition

$$p \Rightarrow (p \vee q)$$

Conjunctive Simplification

$$(p \wedge q) \Rightarrow p \quad (p \wedge q) \Rightarrow q$$

Disjunctive Simplification

$$(p \vee q) \wedge \neg p \Rightarrow q \quad (p \vee q) \wedge \neg q \Rightarrow p$$

Chain Rule

$$(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow (p \rightarrow r)$$

Resolution

$$(\neg p \vee r) \wedge (p \vee q) \Rightarrow (q \vee r)$$

Conditional Equivalence

$$p \rightarrow q \Leftrightarrow \neg p \vee q$$

Biconditional Equivalences

$$(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p) \\ \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$$

Contrapositive

$$p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$$